Examiner only

5. Zinc reacts with sulfuric acid, H_2SO_4, to produce zinc sulfate and hydrogen gas.

(a) Write the formula of zinc sulfate to complete the equation for this reaction. [1]

$$Zn \quad + \quad H_2SO_4 \longrightarrow \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad + \quad H_2$$

(b) Some Year 10 students decided to investigate the rate of this reaction at 20 °C.

They added 0.2 g of zinc to 50 cm^3 of sulfuric acid and measured the volume of hydrogen gas produced every 20 seconds for 160 seconds using a gas syringe.

Their results are shown in the table.

Time (s)	Volume of hydrogen (cm^3)
0	0
20	20
40	37
60	50
80	60
100	66
120	69
140	70
160	70

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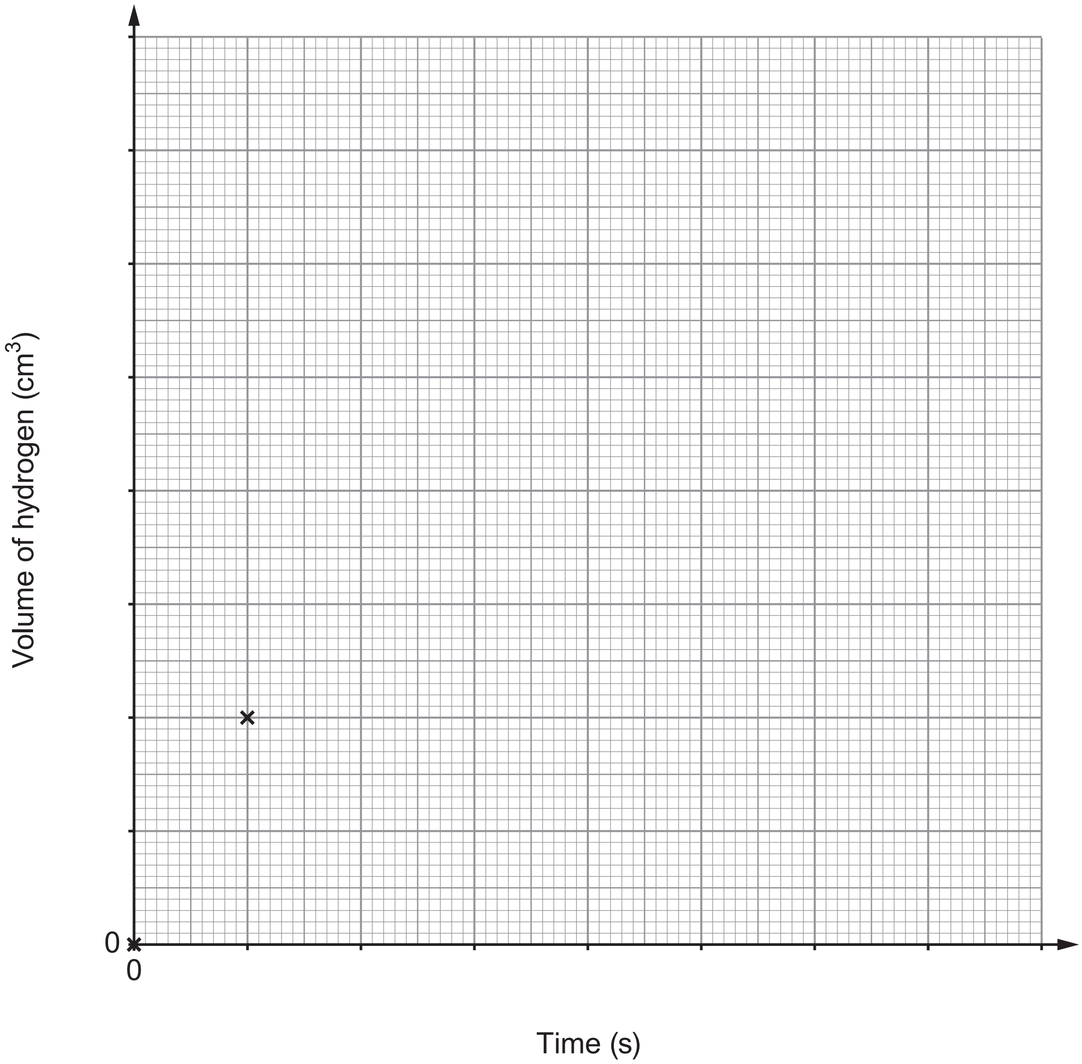
Draw appropriate scales on both axes on the grid.

Plot the volume of hydrogen produced against time and draw a suitable line. [3]

The first two points have been plotted for you.



(c) **On the same grid**, sketch the graph you would expect to obtain if the experiment were repeated at 40 °C using the same mass of zinc and the same volume and concentration of sulfuric acid. [1]

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(d) Use the particle theory to explain what you would expect to happen to the rate of the reaction if the experiment were repeated at 20 °C using sulfuric acid of **lower concentration**. [3]

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(e) Copper is a useful catalyst in the reaction between zinc and sulfuric acid.

State how a catalyst increases the rate of a reaction. [1]

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9